

● EASY

● PROBLEM-SOLVING AND DATA ANALYSIS

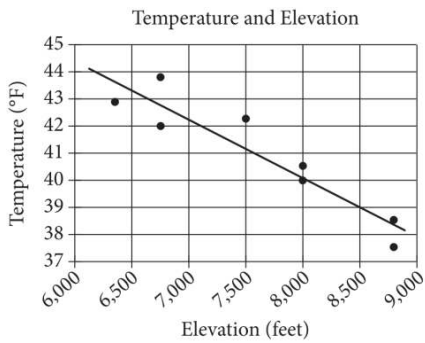
Problem-Solving and Data Analysis

07

10 questions · ~15 min

Q1

PROBLEM-SOLVING AND DATA ANALYSIS



The scatterplot above shows the high temperature on a certain day and the elevation of 8 different locations in the Lake Tahoe Basin. A line of best fit for the data is also shown. What temperature is predicted by the line of best fit for a location in the Lake Tahoe Basin with an elevation of 8,500 feet?

- A. 37°F
- B. 39°F
- C. 41°F
- D. 43°F

Q2

GRID-IN

PROBLEM-SOLVING AND DATA ANALYSIS

How many feet are equivalent to 34 yards? 1 yard = 3 feet

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q3

PROBLEM-SOLVING AND DATA ANALYSIS

What number is 20% greater than 60 ?

- A. 50
- B. 72
- C. 75
- D. 132

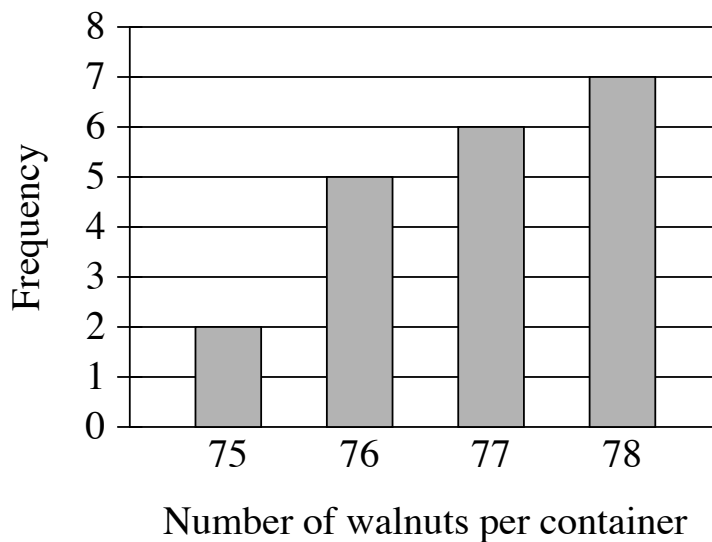
Q4

PROBLEM-SOLVING AND DATA ANALYSIS

A band with 45 members has 11 members who play saxophone. If one band member is selected at random, what is the probability of selecting a band member who plays saxophone?

- A. $\frac{1}{45}$
- B. $\frac{11}{45}$
- C. $\frac{34}{45}$
- D. $\frac{45}{45}$

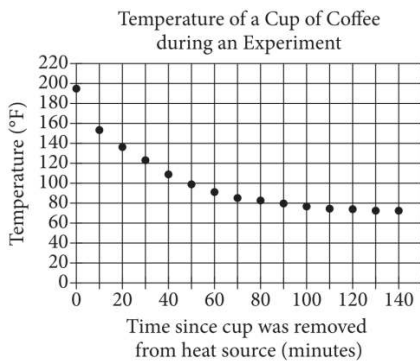
The bar graph shows the distribution of the number of walnuts per container for 20 containers at a grocery store.



- The Frequency data for the 4 bars are as follows:
 - 75 walnuts per container: 2
 - 76 walnuts per container: 5
 - 77 walnuts per container: 6
 - 78 walnuts per container: 7

How many of these containers of walnuts contain exactly 78 walnuts?

- A. 2
- B. 7
- C. 20
- D. 78



In an experiment, a heated cup of coffee is removed from a heat source, and the cup of coffee is then left in a room that is kept at a constant temperature. The graph above shows the temperature, in degrees Fahrenheit ($^{\circ}\text{F}$), of the coffee immediately after being removed from the heat source and at 10-minute intervals thereafter. During which of the following 10-minute intervals does the temperature of the coffee decrease at the greatest average rate?

- A. Between 0 and 10 minutes
- B. Between 30 and 40 minutes
- C. Between 50 and 60 minutes
- D. Between 90 and 100 minutes

Q7

PROBLEM-SOLVING AND DATA ANALYSIS

A kangaroo has a mass of 28 kilograms. What is the kangaroo's mass, in grams?

(1 kilogram = 1,000 grams)

A. 28,000

B. 1,028

C. 972

D. 784

Q8

PROBLEM-SOLVING AND DATA ANALYSIS

Of 900,000 beads, 828,000 are silver. What percentage of the beads are silver?

A. 8%

B. 36%

C. 72%

D. 92%

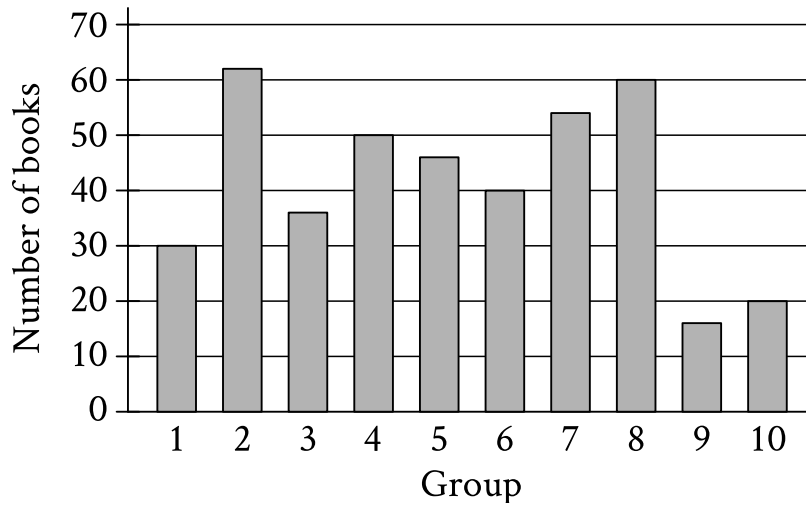
For a particular machine that produces beads, 29 out of every 100 beads it produces have a defect. A bead produced by the machine will be selected at random. What is the probability of selecting a bead that has a defect?

A. $\frac{1}{2,900}$

B. $\frac{1}{29}$

C. $\frac{29}{100}$

D. $\frac{29}{10}$



- The data for the 10 categories are as follows:
 - Group 1: 30
 - Group 2: 62
 - Group 3: 36
 - Group 4: 50
 - Group 5: 46
 - Group 6: 40
 - Group 7: 54
 - Group 8: 60
 - Group 9: 16
 - Group 10: 20

The bar graph shows the distribution of 414 books collected by 10 different groups for a book drive. How many books were collected by group 1?

STUDENT-PRODUCED RESPONSE

☐ 3	☐ 3	☐ 3	☐ 3
☐ 4	☐ 4	☐ 4	☐ 4
☐ 5	☐ 5	☐ 5	☐ 5
☐ 6	☐ 6	☐ 6	☐ 6
☐ 7	☐ 7	☐ 7	☐ 7
☐ 8	☐ 8	☐ 8	☐ 8
☐ 9	☐ 9	☐ 9	☐ 9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.